

Atwood Machine Pre-Lab

Unit 2 • Forces | Completed questions 1–4

Model: A taut, massless string passes over a light, frictionless fixed pulley. The string does not stretch. Air resistance is neglected. The left mass is m_1 ; the right mass is m_2 .

Main relationship: Acceleration depends on the mass difference divided by the total mass.

$$|a| = g |m_2 - m_1| / (m_1 + m_2)$$

The heavier mass accelerates downward and the lighter mass accelerates upward. This describes acceleration; an initial velocity could point the other way.

1. Equal masses

Question: If two equal masses are suspended from either end of a string passing over a light pulley (an Atwood's machine), what kind of motion do you expect to occur? Why?

Answer: If released from rest, both masses remain at rest. Their weights are equal, so there is no unbalanced force driving the system and $a = 0$. The tension balances the weight of each mass: $T = mg$. If the masses are already moving, they continue at constant speed in this ideal model.

2. Changing the masses

Question: For an Atwood's machine, how would you expect the acceleration to change if you (a) Move mass from one side to the other, keeping the total mass constant (b) Gradually increase the mass of both sides?

(a) Keep total mass constant: Acceleration magnitude increases when the mass difference increases. Moving mass from the lighter side to the heavier side increases the acceleration. Moving mass from the heavier side to the lighter side decreases the acceleration until the masses are equal. Further transfer reverses which side accelerates downward and increases the magnitude again.

(b) Add the same amount of mass to each side: The acceleration magnitude decreases if the masses were unequal. The difference in their weights stays the same, but there is more total mass to accelerate. If the masses were equal, the acceleration stays zero.

After adding Δm to each side:

$$|a'| = g |m_2 - m_1| / (m_1 + m_2 + 2\Delta m)$$

Wording note: Part (b) does not specify how mass is added. The answer above assumes equal additions. If both masses are multiplied by the same factor, acceleration is unchanged because the mass difference and total mass scale together. Other unequal additions depend on the resulting ratio.

String constraint & force diagrams

3. Why the accelerations match

Question: Why do the two masses have the same acceleration?

Answer: They have the same **acceleration magnitude**, in opposite vertical directions. A taut string of fixed length requires one mass to move up exactly as far as the other moves down. Their changes in speed therefore have equal magnitudes.

Taking downward as positive on both sides:

$$y_1 + y_2 = \text{constant, so } a_1 + a_2 = 0.$$

Thus $a_1 = -a_2$. The fixed string length causes this relationship; equal tension alone does not explain it.

4. Free-body diagrams

Question: Draw a free-body diagram of the left side mass. Draw another of the right side mass. Include all forces acting on each mass.

Answer: Each hanging mass has exactly two forces: upward tension from the string and downward gravitational force from Earth. The same tension T acts on both masses in this ideal model.



Illustrated case: $m_2 > m_1$, so $m_1g < T < m_2g$.

These arrows show forces **on each mass**. Tension points upward because the string pulls each mass toward the pulley. There is no normal force because neither mass rests on a surface. Acceleration is not an extra force.

Newton's second law: For the illustrated case $m_2 > m_1$, choose upward positive for the left mass and downward positive for the right mass. Both then have the same positive acceleration a .

Left: $T - m_1g = m_1a$

Right: $m_2g - T = m_2a$

Add: $(m_2 - m_1)g = (m_1 + m_2)a$

Source: user-provided Schoology assignment screenshot, September 21, 2026. All four source questions are included; diagrams are redrawn.